

A Python-Based Investigation of Clinical Data and Ultrasound Images for PCOS Diagnosis

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Abstract— PCOS is a common endocrine disorder that impacts women in their reproductive years characterized by irregular menstrual cycles, hyperandrogenism, and polycystic ovaries [3]. The full diagnostic plan is mainly a combination of a pelvic ultrasound besides blood tests of specific parameters that indicate the presence of PCOS. Since PCOS is a hard-to-diagnose widespread hormonal disorder, blood tests, symptoms, and other parameters with the help of a computer can form a new and easy method to diagnose it. This study integrates ultrasound image analysis and numerical data exploration to provide a comprehensive understanding of PCOS.

Ultrasound images were pre-processed using Python-based libraries like OpenCV, with techniques such as cropping, edge detection, pixel intensity analysis, and follicle overlay. These methods highlighted structural differences between normal and PCOS ovaries. Numerical attributes, including BMI, hormonal levels, and lifestyle factors, were analysed.

The findings reveal significant differences in follicle distribution and intensity patterns in ultrasound images of PCOS-affected ovaries compared to normal ones. Statistical analysis further identified BMI, hormonal imbalances, and exercise habits as critical factors linked to PCOS.

This research paves the way for advanced diagnostic tools and contributing to better management of PCOS in clinical practice. Our study employs Teachable Machine to classify PCOS and non-PCOS images, achieving an accuracy of 100%.

Keywords—PCOS, ultrasound image analysis, Python, follicle distribution, BMI, hormonal imbalances

I. INTRODUCTION

Polycystic Ovary Syndrome (PCOS) is one of the most common endocrine disorders affecting women of reproductive age [1]. It is characterized by hormonal imbalances, irregular menstrual cycles, and the presence of multiple ovarian cysts. PCOS affects approximately 8-13% of women worldwide and up to 70% of cases are undiagnosed [4], making it a significant global health issue.[3] This disorder not only disrupts reproductive health but is also associated with a wide range of metabolic complications, including obesity, insulin resistance, and an increased risk of type 2 diabetes.

Despite the high prevalence of PCOS, its diagnosis remains complex and often elusive due to the multifactorial nature of the condition. The symptoms of PCOS can vary widely among individuals and are often subtle, leading to underdiagnosis or delayed diagnosis. Current diagnostic approaches primarily rely on clinical symptoms and hormonal tests. In particular, the variability in clinical presentation and the lack of standardized criteria for diagnosis further complicate the identification of PCOS.

In addition, while ultrasound imaging offers a non-invasive method to detect ovarian abnormalities, it often lacks sufficient precision in distinguishing between PCOS and other conditions with similar symptoms like anaemia. The presence of multiple cysts in the ovaries, a key feature of PCOS, can sometimes be mistaken for other benign conditions, leading to misdiagnosis. The significance of improving PCOS diagnosis and understanding its underlying mechanisms cannot be overstated. By addressing the gaps in current diagnostic approaches, we can enhance our understanding of PCOS and develop more effective strategies for diagnosis, monitoring, and treatment.

Background Information and Problem Definition:

Generally, obstetrician can directly identify PCO follicles. However, extensive time is needed to conduct a series of analysis. Its diagnosis often follows the Rotterdam criteria, requiring at least two of the following features:

1. **Irregular menstrual cycles (oligo-anovulation):**
Highlights hormonal imbalance causing irregular or absent periods.
2. **Hyperandrogenism:**
Elevated androgen levels causing symptoms like acne, hirsutism, or male-pattern baldness, supported by biochemical evidence (e.g. high testosterone levels).
3. **Polycystic ovaries on ultrasound:**
Presence of 12+ small follicles or enlarged ovarian volume.

Despite these criteria, diagnosing PCOS remains challenging due to symptom variability and overlaps with other conditions.

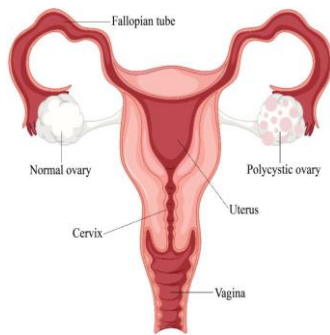


Fig. 1 Polycystic and Normal ovary ^[4]

This study aims to analyze and compare ultrasound images of normal and PCOS-affected ovaries using advanced image processing techniques like edge detection and intensity mapping to identify key features and uncover patterns beyond traditional visual inspection. Additionally, it explores correlations between clinical and lifestyle factors such as BMI, hormonal levels, and menstrual irregularities with PCOS prevalence to better understand risk factors and improve diagnostic accuracy. Lastly, the research seeks to develop an integrated framework combining image analysis and clinical data exploration, enhancing the understanding, diagnosis, and management of PCOS.

II. RELATED WORK

Early Findings on Polycystic Ovary Syndrome (PCOS)

Early studies related to Polycystic Ovary Syndrome (PCOS) were mainly limited to its clinical presentation and hormonal evaluation. Azziz et al. (2004) discussed the diagnostic criteria of PCOS that focused on detection of hyperandrogenism as well as disorders of menstrual cycling. Their work provided the initial basis of diagnosing PCOS, a scheme that the Rotterdam criteria set today as at least one of three; namely oligo- or anovulation, clinical and biochemical signs of hyperandrogenism, and polycystic ovaries as revealed by

ultrasound.^[2] This work set a basis for the understanding of PCOS as a multifaceted disorder, though it was very limited in scope, relying solely on clinical data and often overlooking the potential contribution of lifestyle factors and imaging techniques.^{[2][9]}

Further research studies were then directed to investigate the connection between clinical manifestations and hormonal imbalance. For example, Franks (1995) identified insulin resistance as an integral component in the pathophysiology of PCOS.^{[2][3]} Thus, metabolic factors might have an impact on the clinical expression of the syndrome. Yet, such initial researches did not provide the most holistic datasets which encompassed clinical data as well as imaging, which constrained them to reach any robust conclusions about the interplay of different factors in PCOS.

PCOS diagnosis typically involves assessing irregular menstrual cycles, hyperandrogenism (elevated androgens causing acne, hirsutism, etc.) and ovarian ultrasound findings of multiple cysts. Ultrasound imaging is central but relies heavily on observer expertise, leading to potential errors. To address these challenges, researchers have explored automated approaches combining image processing and machine learning.^[2]

Image Processing Techniques:

- Methods include noise reduction, segmentation, and feature extraction.
- Techniques like k-means clustering, median filtering, and active contour models are commonly used.
- Challenges include high computational complexity and labour-intensive tuning for diverse image qualities.^[8]

Machine Learning Approaches:

- Combining image processing with classifiers like SVM (Support Vector Machine) and Decision Trees improves diagnosis accuracy. SVM often outperforms others.
- Ensemble methods (bagging, boosting, stacking) are emerging as promising solutions to enhance prediction accuracy.

Advances in PCOS Detection: Ultrasound Imaging, Machine Learning, and Clinical Integration ^[10]

- 1 **Ultrasound-Based Approaches:** M. Sumathi et al. achieved 99% accuracy in image classification using DarkNet-19. Bedy Purnama et al. reached 82% accuracy with Gabor wavelets and SVM-RBF. Yinhui Deng et al. introduced an object-growing algorithm with 89.4% recognition. Sharvari S. Deshpande combined SVM with morphological and edge detection for 95% accuracy. Sayma Alma Suham et al. used CNN with XGBoost, achieving 99.89% accuracy. Pradeep Bedi et al. proposed AResUNet for multi-modal images, achieving 98% accuracy.^{[2][4][5]}
- 2 **Clinical Data & Hybrid Models:** Ejay Nsugbe enhanced dataset balance through synthetic generation and achieved optimal performance with SVM.

Random Forest and ensemble classifiers also showed high accuracy, reaching up to 98.87% in various studies. Shazia Nasim achieved 100% accuracy by using chi-squared feature selection combined with Gaussian Naive Bayes. Jay Jojo Cheng applied machine learning to ultrasound reports, reaching an accuracy of up to 97.6%.^{[3][7]}

- 3 **Integrated Approaches:** Hybrid models that combine ultrasound and clinical data, such as the study using Rotterdam criteria, offer a more comprehensive diagnostic approach. By integrating both imaging and clinical factors, these models enhance diagnostic accuracy and provide a holistic view of conditions like PCOS, improving the reliability of assessments and treatment recommendations.^{[2][3]}

This research highlights the advancements in PCOS detection, emphasizing the need for standardized, integrated approaches using hybrid datasets for better prediction.^{[3][7]}

III.DATASET DESCRIPTION

This study uses two primary datasets, each contributing distinct information to the analysis and understanding of Polycystic Ovary Syndrome (PCOS)

The study utilizes two datasets: a clinical dataset and an ultrasound image dataset. The clinical dataset comprises 541 samples with 43 attributes, covering demographics, hormones, lifestyle, reproductive health, and ultrasound features. It includes key indicators such as BMI, cycle length, hormonal levels, pregnancy history, and follicle data to aid in PCOS detection and analysis. The ultrasound image dataset consists of 2000 images depicting both normal and PCOS-affected ovaries, sourced from Kaggle. These images serve as a crucial resource for visual analysis, helping identify structural markers and differences between healthy and cystic ovaries.

IV. METHODOLOGY

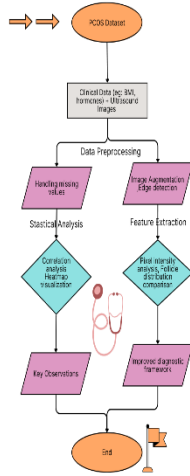


Fig2. Methodology

This study utilizes **Python, NumPy, Pandas, Matplotlib, Seaborn, and OpenCV** for data processing, analysis, and visualization.^{[11][12]}

Step 1: Read and Explore CSV File

Objective: Load and analyse the dataset, focusing on extracting counts of normal and PCOS cases.

Using `pd.read_csv()`, the dataset was loaded and previewed with `head()` to check structure and variable types. `value_counts()` determined the distribution of PCOS cases, while `describe()` provided descriptive statistics for numerical variables.

Step 2: Data Cleaning and Preprocessing

Objective: Prepare the dataset for analysis by handling missing values and outliers.

- **Handling Missing Values:** Checked with `isnull()`, then imputed using mean (numerical) or mode (categorical), or removed if excessive.

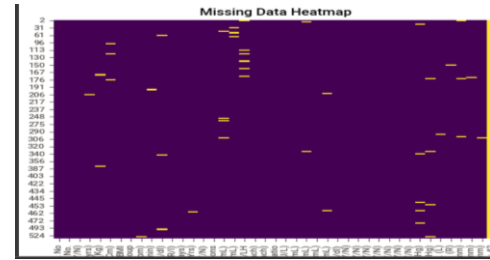


Fig. 3 Missing Data before filling null values

- **Outlier Detection:** Used IQR method; values beyond $Q1 - 1.5IQR$ or $Q3 + 1.5IQR$ were removed.

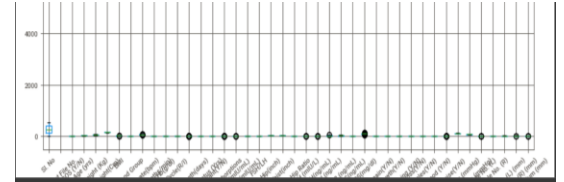


Fig. 4 Outliers Before handling

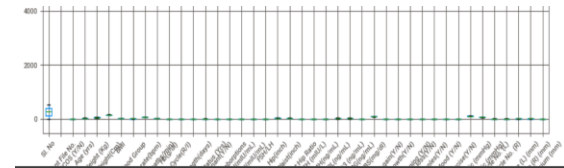


Fig. 5 Outliers after Handling

- **Data Type Conversion:** Ensure that all columns are of the correct data type (ex, categorical variables are converted to the 'category' type) to facilitate analysis.

Step 3: Exploratory Data Analysis (EDA)

Objective: Explore relationships between key variables and visualize the data.

- **Visualizing Distributions:** Histograms and box plots are created for continuous variables (e.g. age, weight, BMI) to visualize their distributions and identify any skewness or outliers.

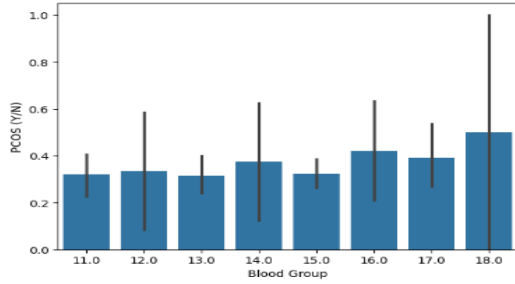


Fig. 6 Histogram of PCOS(Y/N) vs Blood group

- **Correlation Analysis:** A correlation matrix is generated using the `corr()` method to identify relationships between numerical variables.

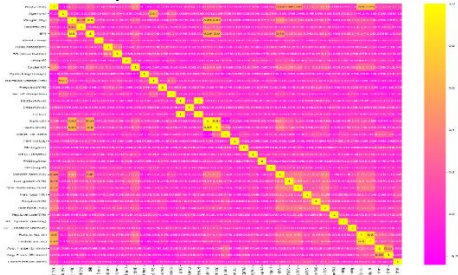


Fig. 7 Correlation Matrix with all attributes

- **Comparative Analysis:** Box plots are used to compare the distributions of key variables (eg: BMI, hormonal levels) between PCOS and non-PCOS patients.

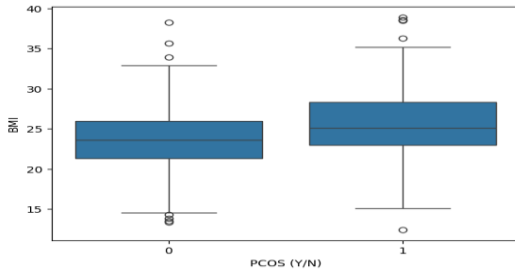


Fig. 8 Box plot of PCOS(Y/N) vs BMI

4. Methodological Steps

Step 1: Read and Explore Zip File using Google drive Mount method

Objective: The clinical dataset is loaded using Pandas, containing patient demographics and clinical measurements. PCOS cases are counted using `value_counts()` to determine the distribution of "Yes" (PCOS) and "No" (non-PCOS) cases. This analysis provides insights into the prevalence of PCOS within the dataset. Understanding this distribution helps in further statistical analysis and model training.

Step 2: Image Preprocessing and Augmentation^[9]

Objective: Prepare ovarian images (normal vs. PCOS) for analysis.

The images undergo preprocessing to isolate the ovarian regions, which involves cropping the images to focus on the relevant structures. This is crucial for ensuring that the analysis is centered on the areas of interest.



Fig. 9 Image Augmentation

To enhance the robustness of the analysis, various image augmentation techniques are applied like flipping and cropping the images, which help to increase the diversity of the dataset.

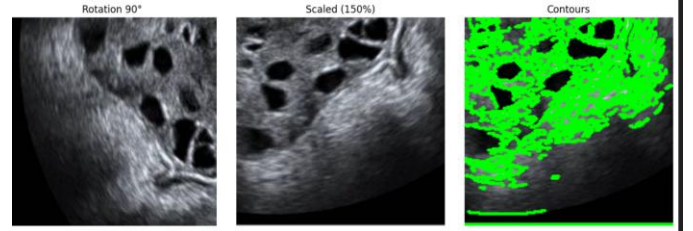


Fig. 10 Image Contouring

Following augmentation, edge detection is performed using the Canny edge detection algorithm. This technique highlights the boundaries of the ovarian structures, making it easier to visualize and analyse the contours of the follicles and cysts present in the images. Contour drawing is then applied to visualize the detected structures, providing a clearer representation of the ovarian morphology.^{[2][3]}

Step 3: Compare Pixel Intensity (Normal vs. PCOS)

Objective: Analyse and compare pixel intensity distributions between normal and PCOS ovaries.

The images are first converted to grayscale to simplify the analysis, as colour information is not necessary for this specific task.

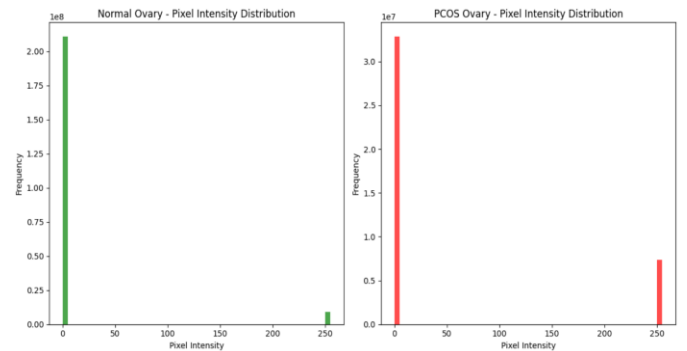


Fig. 11 Histogram of Pixel Intensity

Histograms of pixel intensities are generated for both normal and PCOS ovaries helping to identify any significant differences between the two groups. This gives an idea about the structural differences in ovarian morphology, as variations in intensity can indicate differences in follicle size, density, and overall ovarian health.

Step 4: Image Transformation (Normal and PCOS) ^{[9][10][11]}

Objective: Transform images into different formats and visualize the results.

In this step, the ultrasound images are transformed into various formats, including binary, grayscale, and RGB. Each format serves a different purpose in the analysis.^[7]

- **Binary Images:** These images are useful for highlighting specific features, such as the presence of cysts or follicles, by converting the grayscale images into a binary format based on a defined threshold. This transformation simplifies the analysis by focusing on the presence or absence of features rather than their intensity.
- **Grayscale Images:** Grayscale images are used for general analysis and visualization, as they retain the intensity information while reducing the complexity of the data.

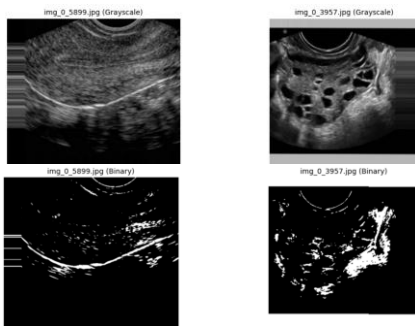


Fig. 12 Binary & Grayscale Conversion

- **RGB Images:** The original RGB images are retained for comparison and visualization purposes, allowing for a comprehensive view of the ovarian structures.

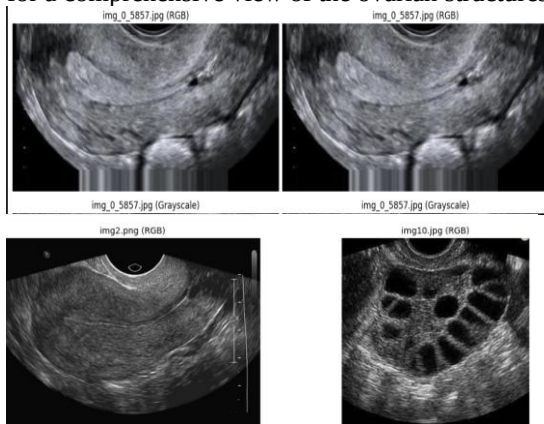


Fig. 13 RGB Conversion

After transforming the images, visualizations are created using Matplotlib to display the different formats side by side. This comparison helps in understanding how the different transformations affect the representation of the ovarian structures.

Step 5: Heatmap Comparison (Normal vs. PCOS)

Objective: Create heatmaps to compare follicle distribution in normal and PCOS ovaries.

Heatmaps visualize follicle distribution in normal and PCOS ovaries by calculating follicle intensities from ultrasound pixel data. Using Seaborn, these heatmaps highlight density variations, aiding in structural comparison. Additionally, PCOS images can be overlaid on normal images for a direct visual contrast.

Step 6: Follicle Overlay and Highlighting

Objective: Highlight follicle points and compare normal vs. PCOS ovaries.

Follicle coordinates are extracted from ultrasound images and overlaid using distinct colors for normal and PCOS ovaries. This visualization highlights differences in distribution and density, revealing the clustering characteristic of PCOS.

Step 7: Convert Image to DataFrame

Objective: Convert image pixel data into a tabular format for analysis.

To facilitate further analysis, the pixel data from the ultrasound images is converted into a tabular format. This involves flattening the pixel intensity values into arrays and creating a DataFrame that includes both the pixel intensities and the corresponding labels (normal or PCOS).

Pixel Intensity Statistics by Class (Normal vs. PCOS):								
Class	count	mean	std	min	25%	50%	75%	max
Normal	660000.0	74.455597	45.624430	0.0	36.0	68.0	103.0	255.0
PCOS	600000.0	70.891578	43.625304	0.0	35.0	67.0	101.0	255.0

Fig.14 Description of data-frame of PCOS and normal images

This tabular representation allows for easier statistical analysis and comparison of pixel intensities between the two groups. It also enables the application of machine learning algorithms, as the data can be easily manipulated and fed into various models for predictive analysis.

Step 8: Visualization and Analysis

Objective: Compare and visualize the differences between an original and an augmented image, helping to understand how augmentations affect the image content.

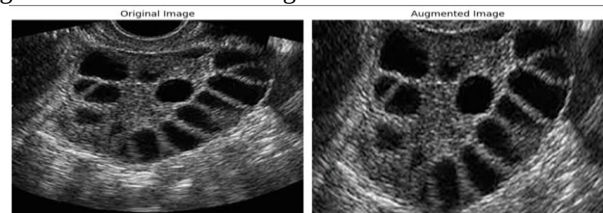


Fig. 15 Original vs Augmented

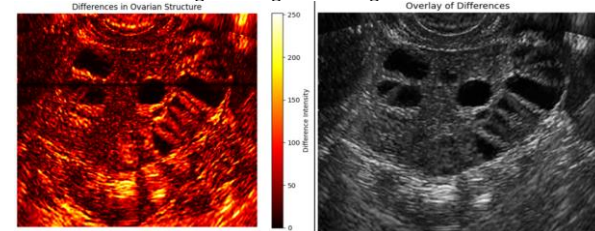


Fig. 16 Difference between original and augmented image using Heatmap

The process involves loading, resizing, and converting images to RGB for display. Three functions—**plot_comparison()**, **visualize_differences()**, and **overlay_differences()**—facilitate side-by-side comparison, heatmap visualization, and transparent overlay of differences. Google's Teachable Machine was used to classify PCOS and non-PCOS images with a CNN-based pre-trained model. The dataset of 2000 images was split 80% for training and 20% for testing. Using transfer learning, the model extracted key features, mapped them to labels, and generated confidence scores for predictions.

V. RESULTS AND DISCUSSIONS

Overview of Experiments:

Experiments analysed ultrasound ovarian images to compare features between normal and PCOS cases. The dataset consisted of 1,000 images reduced from original 2000 for simplicity (500 normal and 500 PCOS). Tools such as Python and OpenCV were used for image preprocessing, including image augmentation, edge detection, contour identification, and pixel intensity analysis to assess follicle distribution and clustering patterns.

Performance Metrics:

Metrics like pixel intensity variation, follicle clustering density, and distribution patterns quantified differences between normal and PCOS cases. Heatmaps and overlays were used for effective visualization.

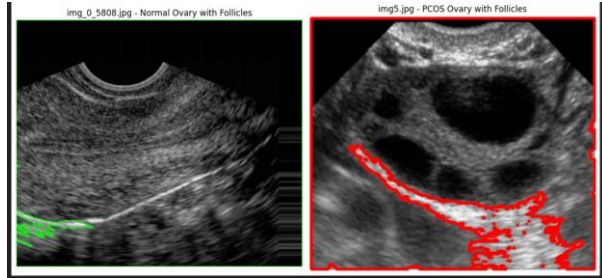


Fig. 17 Follicle Highlighting

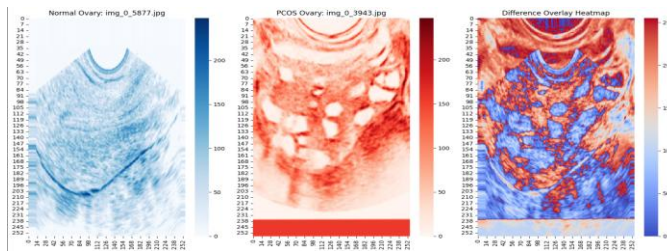


Fig. 18 Heatmap of PCOS vs Normal Ovaries

Analysis of Results:

PCOS ovaries exhibited higher pixel intensity variation and dense follicle clustering, while normal ovaries had lower variation and even distribution. Canny edge detection and contour mapping effectively highlighted follicle structures, though binary thresholding led to some detail loss. The Teachable Machine model achieved 100% accuracy in classifying PCOS and non-PCOS images [14], with confidence

scores near 100% and no misclassifications [4]. Image augmentation improved dataset diversity, while pixel intensity histograms and heatmaps revealed clear distribution differences. Creating a DataFrame enabled deeper statistical analysis, enhancing the interpretation of results.



Fig. 19 Teachable Machine interface showing 100% accuracy in classifying PCOS and non-PCOS images.

Limitations

One limitation of this study is the lack of consistency between clinical and ultrasound data, as they may not always belong to the same individual. Additionally, the dataset includes data from different individuals, which may affect model generalizability. [3][4][13]

VI. CONCLUSIONS

Summary of Work

The study combined image processing, pixel analysis, and transformation techniques to enhance PCOS classification. Heatmaps and follicle overlays highlighted structural differences, while tabular data improved quantitative comparisons. The Teachable Machine model achieved 100% accuracy, validating its effectiveness. This approach could revolutionize PCOS diagnosis, enable non-invasive monitoring, and support ovarian health research.

Future Scope and Potential Developments

Future improvements could include utilizing higher-resolution images and advanced noise reduction techniques to improve accuracy. Adaptive thresholding and segmentation algorithms would enhance follicle and ovarian boundary detection. [3] Expanding the dataset to include more diverse samples and optimizing the methodology for real-time processing would further boost its clinical applicability. Additionally, integrating machine learning techniques like **KNN classifier** could automate follicle detection and classification, improving accuracy and efficiency in diagnosing ovarian disorders such as PCOS. [15] Anemic and PCOS patients experience hormonal fluctuations, making accurate disease classification a significant challenge. [6] Early detection of PCOS is particularly difficult, necessitating the analysis of ovarian ultrasound

images. To address this, we propose developing a **multimodal algorithm** for improved diagnosis and classification.^{[3][13]}

In conclusion, this methodology, enhanced by **Teachable Machine**, shows considerable potential for clinical and research applications in PCOS detection and monitoring.^{[3][4]}

Future advancements in image processing, dataset expansion, and algorithm optimization will increase its utility and effectiveness, providing a robust tool for the early diagnosis and continuous monitoring of PCOS.^{[13][14]}

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